

A floating-potential wall boundary condition with self-consistent

$E \times B$ drifts in JOREK `model600`

Implementation, the SOLPS drift bound, and the zero-sum sheath stabiliser

Abstract

We impose the floating-sheath condition $V_p - V_{\text{wall}} = \Lambda k_B T_e / e$ on the plasma potential at material walls in JOREK `model600`. Because the wall potential then tracks T_e , a tangential electric field appears wherever T_e varies along the wall, and with it a normal $E \times B$ flow — physics that is absent when the potential is held fixed. Making the parallel-flow (Bohm–Chodura) boundary condition consistent with that drift proved to be the hard part: seven successive formulations of the drift-corrected Mach condition all terminated the simulation within 319–649 time steps, always in 2–3 cells at the inner strike point, while the drift-free condition ran indefinitely. The resolution came from a different direction. Porting SOLPS-ITER’s *zero-sum sheath stabiliser* — a term that is identically zero at steady state and therefore cannot change the converged solution — allowed the drift-consistent condition to run past 1100 steps with all diagnostics converging. This note documents the boundary condition, the SOLPS-derived bound that keeps it finite at grazing incidence, the stabiliser, and the evidence.

1 The floating-potential condition

At a material surface the wall charges negative relative to the plasma until the ion and electron fluxes balance. In the local zero-current limit,

$$V_p - V_{\text{wall}} = \Lambda \frac{k_B T_e}{e}, \quad \Lambda \simeq 3, \quad (1)$$

with $\Lambda = \ln \sqrt{m_i / 2\pi m_e}$. Equation (1) is Stangeby’s floating condition written from the plasma side: Stangeby quotes $V_{\text{wall}} - V_p = -\Lambda k_B T_e / e$, the same statement referenced to the plasma instead of the wall.

In JOREK the poloidal velocity is represented through a stream function u with $\mathbf{v}_E = -R \nabla u \times \mathbf{e}_\phi$, and $\Phi = F_0 u$. Condition (1) is therefore *affine* in the evolved variables,

$$u = C_T T_e + C_V V_{\text{wall}}, \quad C_T = \frac{2\Lambda}{a_n}, \quad C_V = \frac{\sqrt{\mu_0 \rho_0}}{F_0}, \quad a_n = \frac{2e F_0 \sqrt{\mu_0 \rho_0}}{m_i}, \quad (2)$$

so it is imposed as a linear constraint on the boundary trace degrees of freedom, exactly, with a constant Jacobian. No Newton iteration, no relaxation and no threshold enter. The condition is applied to the nodal value and to the tangential derivative; both are exact because differentiating a constant-coefficient relation along the boundary is exact degree-of-freedom by degree-of-freedom. Measured residual $|u - C_T T_e - C_V V_{\text{wall}}| \sim 10^{-12}$ V throughout.

The consequence that matters. A wall potential that follows T_e produces a tangential field wherever T_e varies along the wall, hence a normal drift

$$v_{E,n} = -R \frac{\partial u}{\partial \ell} \quad (\text{exact identity, verified against the diagnostic}). \quad (3)$$

With a fixed wall potential $v_{E,n} \equiv 0$ and everything below is dormant. With condition (1) active we measure $|v_{E,n}|$ up to 3×10^4 m s^{−1}, concentrated in two narrow spikes at the inner divertor.

2 Why the drift makes the parallel-flow condition hard

2.1 The condition and its singularity

The drift-compatible Bohm–Chodura condition requires the *total* normal flow at the sheath entrance to be sonic,

$$v_{\parallel} b_n + v_{E,n} = \pm c_s |b_n| \quad \implies \quad v_{\parallel} = \pm c_s - \frac{v_{E,n}}{b_n}. \quad (4)$$

Solved for $v_{\parallel}$ this is an *inversion*, and it is ill posed as $b_n \rightarrow 0$: at grazing incidence no parallel flow can cancel a finite normal $E \times B$ flux. In our equilibrium the demand $v_{E,n}/(c_s b_n)$ has a median of 0.15 across the wall but exceeds unity on 8.3% of it and reaches 3×10^5 at the worst points — a fat tail concentrated where the field is nearly tangent to the surface.

2.2 The SOLPS bound

SOLPS-ITER solves this by bounding the $E \times B$ contribution rather than the geometry (manual 3.0.9, BCMOM=13; multiplier b2stbc_cbc= 1): the poloidal $E \times B$ velocity is “restricted to not exceed $\pm 2c_{s,a}|b_x|$ ”. We adopt the same bound. Critically it is stated in units of c_s , so it introduces no incidence cutoff and no fitted length or angle. We use the *non-marginal* branch, which the manual marks “recommended for cases with drifts”: $v_{\parallel} \geq c_s$ always, never subsonic and never reversed. Written as a one-sided supplement,

$$v_{\parallel} = c_s + \min \left(\max \left(-v_{E,n} / \max(|b_n|, \sin \theta_{\min}), 0 \right), 2c_s \right) \quad (5)$$

so outward drift leaves the parallel flow exactly sonic (extra outward flux is permitted — Bohm is an inequality) and only inward drift is compensated, boundedly. Hence $v_{\parallel} \in [c_s, 3c_s]$. The incidence floor $\theta_{\min}$ re-uses min_sheath_angle= 1° , the same scale that already floors the sheath particle and heat fluxes, so no new parameter is introduced.

Monotone, not smooth. A hard min/max is used deliberately. JOEREK performs one linearised solve per time step, so a branch frozen during a solve must be *exact* within that branch; piecewise-linear clipping satisfies this, whereas a smooth (tanh) saturation has a vanishing tail Jacobian against a finite residual and degenerates into an undamped fixed-point iteration. A non-monotone “graceful surrender” variant, in which the supplement returns to zero when the demand becomes unachievable, was also implemented and tested; it performed *worse* (§4), because across an $E \times B$ spike it replaces one plateau by two peaks flanking a dip, adding tangential structure to $v_{\parallel}$ rather than removing it.

2.3 Consistency of the flux channels

SOLPS requires BCMOM=13 to be used together with BCCON=14 and BCENE/I=15: the momentum, particle and energy conditions must share one collection speed. Two inconsistencies in the JOEREK implementation were found and repaired in the course of this work.

1. The sheath energy transmission was evaluated on the *parallel* normal flux while the volume energy equation convects with the *full* velocity. Since the strong-form volume operator integrates to the surface flux, the two halves of one sheath transmission were being computed on two different flows. The collection is now the total normal flow, $\max(b_n v_{\parallel} + \text{clip}(v_{E,n}), 0)$, floored at zero because a material wall absorbs and never emits.
2. The derivative $\partial c_s / \partial T$ omitted the positivity-map slope corr', although c_s itself is built from corrected temperatures. Below the correction knee this overstates $\partial c_s / \partial T$ by a factor 3 at 0.65 eV and exponentially more below — and the divertor wall in this case sits exactly there.

3 The zero-sum sheath stabiliser

Bounding the momentum condition was necessary but not sufficient: every bounded variant still terminated the run. The resolution came from SOLPS’s own numerics rather than its physics.

SOLPS ships three stabilisers, `b2stbc_stab_coeff_sheath_te`, `_ti` and `_ni`, one for each of the linked drift-compatible conditions, documented as “an additional zero-sum stabilizing term” to suppress target T_e , T_i and n_e oscillations, with a recommended range 1–100 and a default of zero. Reading the reference source (`b2stbc_phys.F`) rather than the manual, all three have the same structure. Using the linearised source decomposition $S = S_0 + S_1 X$,

$$S_1 \text{ --} = (\gamma_{\text{sh}} + \alpha) \Pi, \quad S_0 \text{ +=} \alpha \Pi X_{\text{old}}, \quad \Pi \equiv \text{the physical sheath flux prefactor}, \quad (6)$$

which nets to

$$\boxed{S_{\text{stab}} = \alpha \Pi (X_{\text{old}} - X_{\text{new}})} \quad (7)$$

Three properties follow immediately and are what make this acceptable:

1. **Zero-sum.** The term vanishes identically when $X_{\text{new}} = X_{\text{old}}$, so the *steady state is unchanged*. It is a numerical relaxation of the approach to equilibrium, not a modification of the physics.
2. α is a *dimensionless multiple of the sheath transmission coefficient* — it is added directly alongside γ_{sh} — which is what makes the recommended range 1–100 meaningful.
3. It is pure implicit diagonal: the constant part cancels the residual contribution exactly.

In JOREK the implementation is correspondingly trivial: use $(\gamma_{\text{sh}} + \alpha)$ in the *Jacobian* and γ_{sh} in the *residual*, for the T_i , T_e and ρ boundary rows. With $\alpha = 0$ every previous result is reproduced bit-for-bit.

One caveat, stated explicitly. SOLPS iterates to convergence within a time step, so there the term is purely a convergence aid and the step’s answer is untouched. JOREK takes one linear solve per step, so X_{old} is the previous *time step* and the term also damps the physical transient. The steady state remains unaffected; transients are slowed. For a simulation whose target is a stationary divertor solution this is acceptable, but it should be remembered when interpreting time-dependent behaviour.

Where it acts. Note that SOLPS provides *no* such stabiliser for the momentum condition BCMOM=13. The reference code damps the transport rows and leaves the parallel-flow condition alone — which matches the observed failure mode, in which n_e and T_e lose positivity together in the same boundary cells.

4 Evidence

All runs use the same equilibrium (ASDEX Upgrade, `mode1600`, axisymmetric, bicubic elements, separate T_i/T_e , kinetic neutrals) and identical numerical settings apart from the stated change.

The failure mode was identical throughout: a single-step loss of positivity in 2–3 cells at the inner strike point, with n_e and T_e going negative *together* — the signature of a shared factor, namely the parallel compression $\nabla \cdot (\rho v_{\parallel} \mathbf{b})$, which appears with the same coefficient in the density and both temperature equations. No precursor: the preceding outputs are flat and healthy.

In the stabilised run every diagnostic trends the right way, which no previous drift-enabled run did: minimum boundary density constant to four significant figures; $|v_{E,n}|$ and the applied drift correction both decreasing; and the successive decrements of the minimum wall temperature *shrinking* ($-1.4, -1.2, -1.1, -0.8 \times 10^{-3}$ per output), i.e. an asymptotic approach rather than a collapse.

Configuration	Terminates at step
Fixed wall potential ($\Lambda = 0$, no drift)	runs to time limit
Floating potential, drift term <i>disabled</i>	runs to time limit
Drift term, unbounded (legacy)	605
Drift term, unbounded, with corrected energy flux	606
Drift weighted by incidence	624
Drift smoothly saturated (tanh)	319
Drift clipped, marginal branch (reversal permitted)	466
Drift clipped, non-marginal, one-sided (Eq. 5)	612
+ temperature floor on the potential row	616
+ total-flow energy transmission	610
+ all Jacobian repairs	649
+ achievability weighting (non-monotone variant)	550
+ temperature shock capturing	684
+ zero-sum sheath stabiliser (Eq. 7)	> 1100, converging

Table 1: Seven distinct formulations of the drift-corrected momentum condition all terminate; the stabiliser, which does not touch that condition at all, does not.

5 Suggested figures

1. **Φ and $3T_e$ along the wall, overlaid.** Demonstrates the condition is imposed exactly, and shows at a glance where the potential structure comes from. One line.
2. **n_e and T_e maps at the inner strike point, at the crash step, negatives highlighted.** Two panels. Establishes that the failure is 2–3 cells and that both fields lose positivity in the *same* cells — the single most diagnostic observation.
3. **Wall profiles of c_s , $|v_{E,n}|$ and b_n versus poloidal angle.** Shows the $\sim 150\times$ temperature contrast across ~ 0.3 rad and that the drift spikes coincide with the temperature fronts. Plot $v_{\parallel}$ itself, *not* $v_{\parallel}b_n$: the latter hides a large parallel velocity wherever the field grazes the surface.
4. **Time traces, stabilised versus unstabilised, on one axis:** minimum wall T_e , minimum wall n_e , and the applied drift correction. This is the result — flat-then-dead against converging.
5. **Histogram of the demand $v_{E,n}/(c_s b_n)$ over the wall, log axis, with the $2c_s$ bound marked.** Justifies the bound quantitatively: median 0.15, but a tail reaching 10^5 .
6. **Optional, for the numerics-minded:** minimum wall T_e versus step for $\alpha = 0$ and $\alpha = 100$, with the steady-state values annotated as equal. Makes the zero-sum property visible rather than merely asserted.

6 Status and open points

Settled. The floating condition itself is exact and is not in question. The bound of Eq. (5) is SOLPS’s, in c_s units, and introduces no fitted parameter. The two flux inconsistencies of §2.3 were genuine errors and are repaired. The stabiliser is zero-sum, so whatever stationary state the run reaches is the answer it would have reached without it.

Open.

- Confirmation that the stabilised run reaches a stationary state rather than merely a slower transient. The zero-sum property guarantees the endpoint, not that it is attained.
- α is a genuine numerical parameter, default zero. It should be reported with any result, and ideally shown not to change the converged solution (figure 6).
- Kinetic recycling still uses the parallel flux alone while the fluid wall now uses the total flow; particle conservation across the fluid–kinetic interface is therefore inconsistent at the strike point. This is independent of everything above and should be addressed.
- The alternative weak (boundary-integral) formulation of the momentum condition remains untested: in the one run attempted the $E \times B$ diagnostic read identically zero, which is unexplained and may indicate a defect in that path.
- The underlying divertor state — a ~ 0.25 eV trough between ~ 40 eV shoulders over one to two elements — lies in the regime where this model has no volumetric energy sink and the atomic physics is gated off. The boundary condition faithfully amplifies that structure; it does not create it.